AI Student Tutor – Solution Design Document

1. Executive Summary

Purpose: Develop a modular, multilingual, multimodal AI tutor for students, capable of providing grounded answers, step-by-step explanations, flashcards, quizzes, content generation (videos/images), progress tracking, and modular model selection.

Scope: Classes (multi-class), Subjects (multi-subject), Chapters, Features (Q&A;, PDF viewer, flashcards, step-by-step, translation, quizzes, videos/images), Deployment: local Windows 11 CPU-based system.

Target Audience: Students, educators, and curriculum administrators.

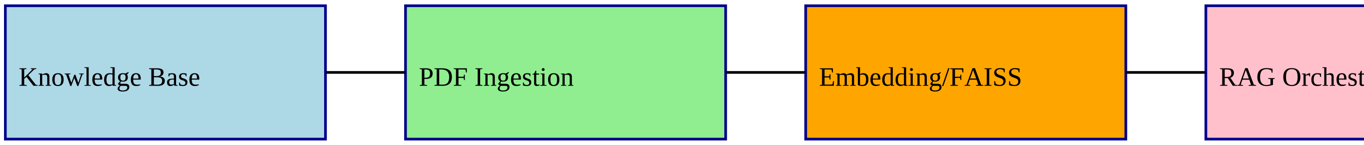
2. Functional Requirements

ID	Requirement	Description
FR1	Multi-class / subject / chapter support	Student selects class → subject → chapter
FR2	PDF ingestion	Import textbooks, notes, question papers with text + images
FR3	Text retrieval	Retrieve relevant text chunks per query
FR4	Image comprehension	Extract diagrams/images and provide explanations
FR5	Multilingual support	Detect Hindi/Sanskrit/English; translate to English
FR6	Step-by-step explanations	Sequential guidance for Math, Physics, Chemistry problems
FR7	Flashcards generation	Text + image flashcards per chapter
FR8	Question paper summary	Topic/difficulty summary for past papers
FR9	Knowledge testing	Generate quizzes (MCQ, short answers, fill-in-the-blank)
FR10	Content generation	Short videos & image-based study aids
FR11	Model selection & modularity	Add new models; student selects model per query
FR12	Save & retrieve results	Store answers, flashcards, videos tagged by class/subject/chapter
FR13	Progress tracker	Dashboard with chapters completed, quizzes, flashcards reviewed
FR14	PDF viewer	Inline display of PDFs with image highlighting

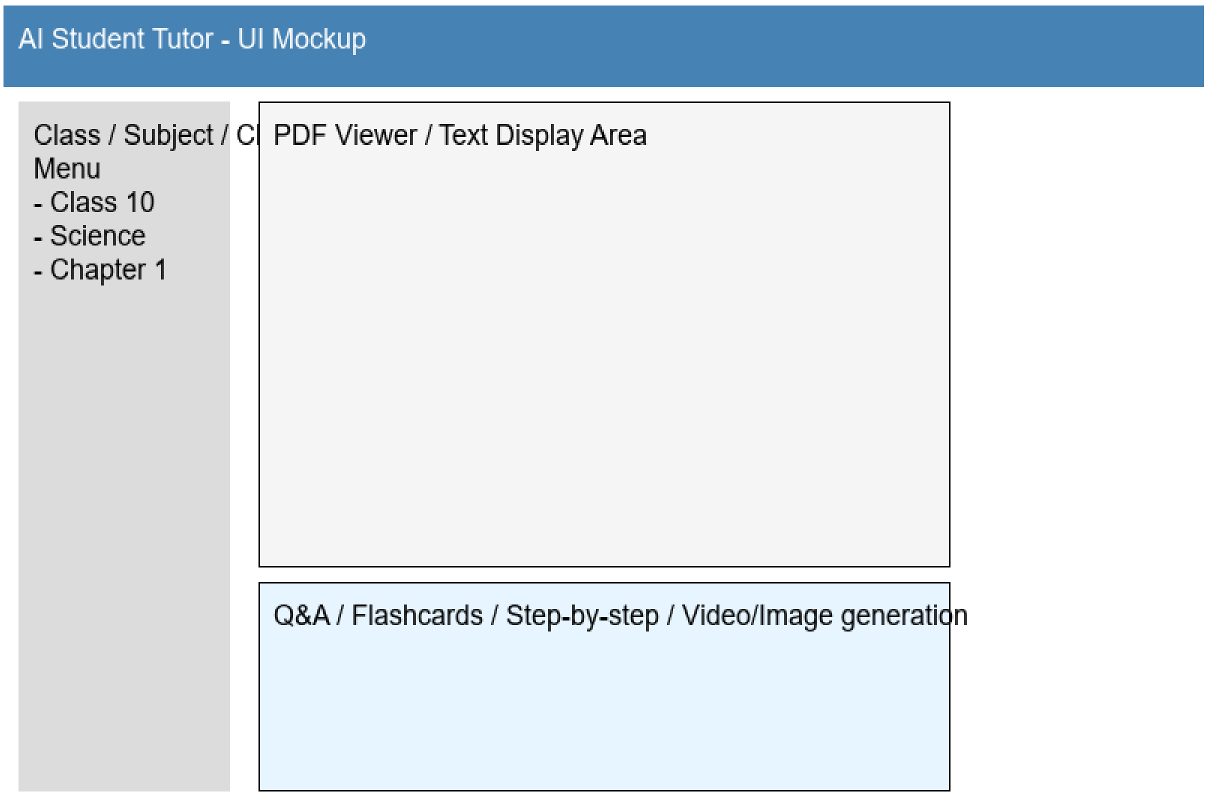
3. Non-Functional Requirements

ID	Requirement	Description
NFR1	Performance	Text Q&A <6s; step-by-step <8s; video/image generation <30s
NFR2	Scalability	Add classes, subjects, PDFs, or models without code changes
NFR3	Availability	Offline-capable local deployment
NFR4	Reliability	Persistent storage for FAISS, saved results
NFR5	Usability	User-friendly GUI, PDF viewer, progress dashboard
NFR6	Maintainability	Modular backend, model-agnostic pipelines
NFR7	Security	Local execution; optional access control for student data
NFR8	Accuracy	Grounded answers with citations, verified translations, step-by-step correctness

4. Architecture Diagram (Vector Flowchart)



5. UI Mockups



6. System / Hardware Requirements

Component	Requirement
Hardware	Windows 11, i5 CPU, 16GB RAM, 250GB storage minimum
Backend Software	Python 3.10+, FAISS, PyPDF2/pdfplumber, BLIP-2 / OpenCLIP
LLM	LLaMA 2 7B Chat Q4_K (CPU quantized), Math-specialized, Hindi/Sanskrit models
Frontend Software	PyQt / Tkinter / Electron
Optional	GPU for faster video/image generation

7. Optional Enhancements

- Precomputed flashcards & summaries
- Highlight images/diagrams in PDF viewer
- Asynchronous video/image generation
- Admin-configurable model selection per subject/language
- Line-by-line translation for Hindi/Sanskrit
- Analytics in progress tracker (heatmaps, chapter completion, quiz performance)

8. Key Benefits

- Modular, extensible architecture
- Grounded, accurate answers with citations
- Step-by-step guidance for problem-solving subjects
- Multilingual support with translation
- Flashcards, quizzes, videos, infographics for active learning
- Progress tracking & analytics
- Offline-capable, suitable for Windows 11 i5 setup